

Composition Pair B1.11 + B1.14 — The Sequencing Pair Where Directed Selection Improvement of Alternative Entities Creates Functional Obsolescence Conditions That Death Governance Addresses, and Death Governance Provides the Governed Closure for Entities Whose Operational Territory Has Been Superseded Through Directed Selection

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to articulate, in operational form, a composition pair between two named architectural commitments — death governance (B1.11) and directed selection (B1.14) — so that their sequencing relationship is explicit and defensible as prior art.

Abstract

The CKS architecture introduced in the source paper (Li, April 2026) names directed selection as the mechanism by which human-governed DNA modification improves an entity’s orchestration substrates along a predictable trajectory. The same architecture names death as the governed lifecycle closure for entities that have reached functional obsolescence or whose lineage has been superseded. This note formalizes the sequencing relationship between these two commitments: directed selection (B1.14) may create the conditions that death governance (B1.11) addresses. When directed selection improves alternative or replacement entities sufficiently, two death-triggering conditions can arise — functional obsolescence (B2.52), where the improved alternatives have taken over an existing entity’s operational territory, and lineage supersession (B2.53), where a superior successor has outperformed the parent. Without B1.11, entities made obsolete by directed selection improvement accumulate in the deployment without governed closure. Without B1.14, the specific improvement-driven pathway to functional obsolescence does not produce candidates for death governance, though death governance remains functional for other reasons. Together, the pair describes a deployment evolution lifecycle: entities are born, directed selection improves them and their alternatives, improvement-driven obsolescence conditions arise, death governance provides governed closure, and the deployment’s entity population is refreshed. The note formalizes pair identification, the enabling direction of the sequencing relationship, mutual dependency, co-presence requirement, operational integration, detection, and failure modes.

1. Pair Identification

Pair: B1.11 (death — governed lifecycle closure) + B1.14 (directed selection — human-governed DNA modification)

Relationship type: Sequencing pair. B1.14 may trigger the conditions that B1.11 addresses. B1.11 provides the governed response to the outcomes B1.14 produces.

Parent commitments. B1.11 derives from the source paper’s treatment of death as governed lifecycle retirement with two distinct types: functional obsolescence (B2.52) and lineage supersession (B2.53), with archival reactivatability per B2.54 for the lineage supersession type. B1.14 derives from the source paper’s treatment of directed selection as human-governed orchestration substrate updates — the evolution mechanism with predictable trajectory, evaluated by intent and outcome together (source paper §7.2).

Why a sequencing pair, not a standard composition pair. Most composition pairs express concepts that co-exist and mutually constrain each other at the same moment. This pair operates in sequence: B1.14 acts first (improving entities over time), and the conditions B1.14 creates feed forward to B1.11 (which responds with governed closure when those conditions cross the obsolescence threshold). The sequencing relationship does not eliminate mutual dependency — it specifies the causal direction of the dependency.

What the pair formalizes. The pair formalizes that a complete governed deployment requires not only a mechanism for improving its entities through directed selection, but also a mechanism for governed closure of the entities that directed selection improvement has rendered obsolete. Without both, the deployment either does not improve (if B1.14 is absent) or improves indefinitely while accumulating the outdated entities improvement has left behind (if B1.11 is absent).

2. The Enabling Direction: How B1.14 Creates Conditions for B1.11

Directed selection per B1.14 is the human-governed process by which orchestration substrates — the DNA layer of entities — are updated along a trajectory that governance can evaluate by intent and outcome together. When directed selection is applied to new, alternative, or replacement entities, it can improve them relative to existing entities along dimensions that matter to the deployment’s purpose.

Functional obsolescence via directed selection (B2.52). The first death-triggering condition arises when directed selection has improved the deployment such that an existing entity’s operational territory is no longer needed. The improved alternatives or reorganized entities now cover what the existing entity once handled; its function has been superseded by improvement rather than by external requirement changes or governance decisions to consolidate. The entity becomes functionally obsolete. The source paper commits to functional obsolescence as genuine deletion: the entity’s substrate content is removed and resources are released, because nothing architecturally requires preservation of an entity whose function is now handled better elsewhere.

Without B1.11, a functionally obsolete entity — one whose territory has been taken over by directed-selection improvements — remains in the deployment without governed closure. It is not formally closed, not archived, not removed. Its substrate resources remain occupied. Governance has no mechanism to clean up what directed selection’s improvements have made redundant. The deployment

accumulates functionally obsolete entities in proportion to how actively directed selection has been applied.

Lineage supersession via directed selection (B2.53). The second death-triggering condition arises when directed selection on a mating offspring or alternative entity has produced a successor that is demonstrably superior along the trajectory governance was pursuing. The parent entity's lineage is superseded — not because the parent was inadequate, but because directed selection has produced something better. The source paper commits to lineage supersession as retirement with archival: the superseded entity's substrate content remains addressable, because it carries the accumulated DNA evolution history that the successor inherited.

Without B1.11, an entity whose lineage has been superseded through directed selection improvement has no governed closure mechanism. It remains in the deployment alongside its superior successor, with no formal status change, no archival, and no lineage closure. Governance cannot mark the supersession event in the substrate. The historical record of which entity superseded which becomes unaddressable through governed substrate content.

B1.14 as cause; B1.11 as response. The enabling direction is unambiguous: directed selection per B1.14 may create the conditions that death governance per B1.11 addresses. B1.14 is the cause of the conditions; B1.11 is the governed response. The sequencing is: B1.14 improves → improvement creates obsolescence conditions → B1.11 provides closure. This direction does not preclude B1.11 operating for reasons unrelated to B1.14 — external scope changes, governance decisions to consolidate entities, or retirement decisions based on factors other than improvement. But the specifically improvement-driven pathway to functional obsolescence and lineage supersession runs through B1.14.

3. The Governed Response: How B1.11 Addresses B1.14 Outcomes

When directed selection per B1.14 has created functional obsolescence or lineage supersession conditions, death governance per B1.11 provides the governed response through four mechanisms.

Death pattern identification. Governance distinguishes whether the condition is functional obsolescence (B2.52) — where the entity's function is genuinely no longer needed because improved alternatives have taken over — or lineage supersession (B2.53) — where the entity's lineage has been superseded by a directed-selection-improved successor. The distinction matters architecturally because the two types have different governed outcomes: functional obsolescence commits to deletion; lineage supersession commits to archival with retained addressability.

Governance decision per A1.01. Human governance decides to close the entity. The governance commitment per A1.01 applies: humans hold the right to inspect, modify, and override at any time. The decision to close is an exercise of governance authority, not an automated outcome of the directed selection events that created the obsolescence condition. The causal chain runs from B1.14 improvement to obsolescence condition, but the closure decision is a human governance act, not an automatic consequence.

Archival state per B2.54 for lineage supersession cases. Where the death type is lineage supersession, the entity is archived rather than deleted. Its accumulated operational knowledge and the DNA evolution

history that directed selection built up are preserved. The source paper's commitment to archival reactivatability means the archived entity is operationally retired but substrate-addressable — the history of what directed selection produced in that entity remains inspectable by governance.

Operational responsibility transfer per B2.55. Where any aspects or Selves referenced the closing entity, the references are resolved. The deployment's entity population is updated to reflect that the closed entity's functional territory is now held by the improved alternatives that directed selection created.

4. Mutual Dependency

The sequencing direction does not collapse into a one-way dependency. The pair is mutually dependent in two respects.

B1.14 provides content for the improvement-driven pathway to B1.11 functional obsolescence.

Functional obsolescence per B2.52 requires that there exist improved alternatives whose presence renders the existing entity's territory unnecessary. Directed selection per B1.14 is the mechanism that produces those improved alternatives. Without B1.14, alternatives do not improve through directed selection, and the specifically improvement-driven pathway to functional obsolescence does not produce candidates. Other pathways to death — external requirement changes, governance decisions to consolidate, scope reduction — remain operational without B1.14. But the pathway the pair jointly describes — directed selection improvement rendering an existing entity's territory obsolete — requires B1.14 as its content supplier.

B1.11 provides governed closure for B1.14 consequences. Directed selection per B1.14 may improve the deployment continuously. Over time, earlier entity versions and superseded entities accumulate as the products of improvement cycles that have moved on. Without B1.11, the deployment fills with entities that directed selection has made outdated — entities whose DNA was the predecessor of what B1.14 subsequently improved. B1.11 provides the governed mechanism to close what B1.14 obsoletes. Without this closure, the deployment's entity population is the cumulative residue of every improvement cycle, not a governed, maintained population.

The asymmetry. The mutual dependency is asymmetric in degree. B1.14 strictly requires B1.11 for the deployment to remain governed — every directed selection cycle that produces improvements also produces obsolescence candidates, and without B1.11 those candidates accumulate indefinitely. B1.11 does not strictly require B1.14 for basic death governance functionality — it can operate on death candidates produced by other causes. The asymmetry does not dissolve the dependency; it specifies that B1.14 creates a stronger architectural requirement for B1.11 than B1.11 creates for B1.14.

5. Directed Selection as Deployment Restructurer

A secondary framing the pair makes visible is directed selection's dual role: when B1.14 is applied to create significantly improved replacement entities, it acts simultaneously as entity improver and as deployment restructurer.

As entity improver, directed selection modifies the new entity's DNA layer — its orchestration substrates — along a trajectory governance can evaluate. The new entity becomes more capable, better aligned to its purpose, or more efficient in its operation. This is directed selection's primary identity in the source paper.

As deployment restructurer, the same directed selection events create the conditions that obsolete the old entity. The improved new entity expands to cover the old entity's functional territory. The deployment's entity population is being restructured: the new entity's territory has grown to encompass what the old entity once held. B1.11 provides the governed restructuring endpoint — the old entity is closed per B2.52 as the improved entity takes over its functional territory, and the closure is a governed act rather than a silent deletion or an indefinite accumulation.

Without B1.11, directed selection can act as entity improver but cannot cleanly act as deployment restructurer. The restructuring it begins — by creating improved alternatives — has no governed completion mechanism. The old entity and the improved replacement coexist in the deployment without formal status differentiation. Governance cannot mark the restructuring as complete.

6. The Deployment Evolution Lifecycle

The pair together describes a deployment evolution lifecycle that neither commitment specifies alone.

Birth (per the lifecycle primitives of the source paper §6.2) creates new entities. Directed selection per B1.14 improves those entities and their alternatives over time — adjusting DNA layer content, refining orchestration substrates, and producing successors through mating combined with directed improvement. Sufficiently improved alternatives render some existing entities functionally obsolete per B2.52, or produce superior successors that supersede lineages per B2.53. Death governance per B1.11 closes the obsolete entities through governed decisions. The deployment's entity population is refreshed: the accumulation of outdated entities is governed out, and the active population reflects the current state of directed selection improvement rather than its entire cumulative history.

Without B1.14, entities never become improvement-obsolete through this mechanism. They may become obsolete through other causes, but the continuous directed selection improvement pathway — the primary driver of ongoing deployment evolution in the source paper's architecture — does not produce the obsolescence candidates that give the lifecycle its refresh mechanism.

Without B1.11, obsolete entities accumulate indefinitely. Each directed selection cycle that produces improvements also produces entities whose territories have been superseded, and those entities have no governed exit from the deployment. The deployment grows with each improvement cycle rather than refreshing.

The lifecycle is not self-sustaining without both commitments. B1.14 generates the evolutionary pressure; B1.11 governs the response to that pressure. The pair holds the lifecycle in governed shape.

7. Co-Presence Requirement

B1.14 without B1.11 (directed selection without death governance). Directed selection continuously improves entities and their alternatives. Older versions and superseded entities accumulate in the

deployment without governed closure. The deployment grows indefinitely with entities that directed selection has rendered obsolete. Governance cannot clean up what B1.14 obsoletes; there is no governed mechanism by which the restructuring directed selection begins can be completed. The deployment's entity population becomes an unsorted accumulation of every improvement cycle rather than a governed, current-state population.

B1.11 without B1.14 (death governance without directed selection as improvement driver). Death governance is fully functional for causes other than directed selection improvement — external scope changes, governance consolidation decisions, requirement changes. The improvement-driven functional obsolescence pathway does not produce candidates, because no mechanism is improving alternative entities along a directed trajectory. The deployment may still close entities, but the continuous evolution cycle the pair jointly describes — improvement producing obsolescence producing closure producing a refreshed population — does not operate.

Full co-presence. Directed selection improves the deployment's entities and their alternatives. Governance reviews periodically assess whether any entities are now functionally obsolete per B2.52 because improved alternatives have superseded their territory, or lineage-superseded per B2.53 because directed selection has produced demonstrably superior successors. When obsolescence is identified, death governance per B1.11 is initiated. The deployment maintains a governed, refreshed entity population.

8. Operational Integration

The pair's operational integration follows a periodic governance review pattern.

Directed selection events per B1.14 improve entities over time. Each such event modifies DNA layer content and advances the directed trajectory of one or more entities. Governance records the improvement event as substrate content.

Governance reviews — triggered periodically or at significant directed selection milestones — assess whether any entities are now functionally obsolete per B2.52 or lineage-superseded per B2.53. The review asks: has directed selection improved the deployment such that an existing entity's operational territory is now covered by superior alternatives? Has a directed selection event produced a successor that supersedes the parent lineage?

When the assessment identifies obsolescence, death governance per B1.11 is initiated. The governance decision is recorded as substrate content. For functional obsolescence cases, the entity's substrate content is removed and resources are released. For lineage supersession cases, the entity is archived per B2.54 with its directed selection history preserved. Lineage closure per B2.43 closes the entity's lineage chain. Operational responsibility transfer per B2.55 resolves any references.

The deployment's entity population reflects the outcome: the entities whose territories directed selection has superseded are governed out; the entities that directed selection has improved are active. The improvement cycle can continue without accumulating its own residue.

9. Detection

Two verification points identify whether the pair is correctly instantiated.

B2.55 death governance verification (B1.11 side). Were death patterns correctly identified and differentiated — functional obsolescence versus lineage supersession? Was the governance decision recorded as substrate content? Was the correct outcome applied — deletion for functional obsolescence, archival for lineage supersession? Was lineage closure completed? Were operational references resolved?

B2.72 directed selection verification (B1.14 side). Are directed selection improvements creating functional obsolescence conditions that are assessed per B2.52? Is there a governance process for reviewing whether directed selection improvements have superseded existing entities' operational territories?

Pair completeness check. Is there a governance process for reviewing whether directed selection improvements have created functional obsolescence or lineage supersession conditions? If directed selection events are recorded but no corresponding governance review for obsolescence exists, the pair is incomplete: B1.14 is operating but its outputs are not feeding B1.11. The improvement cycle generates residue that accumulates without governed closure.

10. Failure Modes

Silent Death (B3.11) when B1.11 is absent. Obsolete entities are simply deleted without governed closure, without archival of directed selection history, without lineage closure, and without operational reference resolution. The deployment loses the audit trail of which entities directed selection improvement superseded and when. Lineage histories become unaddressable. The governance commitment over the deletion decision — the requirement that humans hold authority over what gets closed under what criteria — is absent. Silent deletion may appear functionally equivalent to death governance for routine cases, but eliminates the architectural properties that make directed selection history inspectable and decisions traceable.

Evolution Stasis (B3.24) when B1.14 is absent. No directed selection mechanism improves the deployment's entities along a governed trajectory. Entities do not become improvement-obsolete because no improvement mechanism is producing alternatives that could supersede their territories. Death governance per B1.11 remains operational for external-cause deaths, but the improvement-driven evolution cycle does not operate. The deployment is static with respect to the directed improvement mechanism the source paper names as the primary driver of predictable, trajectory-governed evolution. Death governance has no improvement-driven candidates to process.

Source Paper

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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